



Urban Informatics

Assignment - Week 3

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 16

Total marks: (12 X 1) + (4 X 2) = 20 marks

MCQ/MSQ Question

QUESTION 1: [1 mark]

Why is MySQL classified as a relational database system?

- a) It stores data only in hierarchical structures
- b) It enforces relationships between tables using keys
- c) It supports only spatial datasets
- d) It does not require structured queries

Correct Answer: b. It enforces relationships between tables using keys

Detailed explanation: Refer to lecture 11.

QUESTION 2 : [1 mark]

Which MySQL component primarily simplifies database interaction for users unfamiliar with command-line syntax?

- a) MySQL Server
- b) MySQL Storage Engine
- c) MySQL Workbench
- d) MySQL Connector

Correct Answer: c. MySQL Workbench

Detailed explanation: Refer to lecture 11.

QUESTION 3 : [1 mark]

In a SELECT query containing both WHERE and GROUP BY clauses, which operation is executed first?

- a) Group formation
- b) Row filtering
- c) Result sorting
- d) Aggregate calculation

Correct Answer: b. Row filtering

Detailed explanation: Refer to lecture 13.



QUESTION 4 : [1 marks]

Which SQL technique allows results from one query to be used as input for another query?

- a) Join
- b) Subquery/Nested Subquery
- c) Index
- d) View

Correct Answer: b. Subquery/Nested Subquery

Detailed explanation: Refer to lecture 14.

QUESTION 5 : [2 mark]

Match the following Advanced Database Concepts:

Column I	Column II
P. Index	i. Groups multiple SQL statements into a reusable executable block
Q. View	ii. Improves query performance by speeding up data retrieval
R. Stored Procedure	iii. Ensures atomicity and consistency of database operations
S. Transaction management	iv. Virtual table created from a SELECT query without storing data

- a) P–ii, Q–iv, R–i, S–iii
- b) P–iv, Q–ii, R–iii, S–i
- c) P–ii, Q–i, R–iv, S–iii
- d) P–iii, Q–iv, R–ii, S–i

Correct Answer: a. P–ii, Q–iv, R–i, S–iii

Detailed explanation: Refer to lecture 14.

QUESTION 6 : [1 mark]

Which characteristic best differentiates API-based data access from direct database access?

- a) APIs store data permanently and make sures they are ACID compliant
- b) APIs abstract internal database structure and expose data via controlled interfaces
- c) APIs require SQL from clients so that data can be returned
- d) APIs eliminate authentication

Correct Answer: b. APIs abstract internal database structure and expose data via controlled interfaces

Detailed Solution: Refer to lecture 15.



QUESTION 7 : [1 mark]

Which technology enables real-time updates in traffic monitoring dashboards?

- a) REST polling
- b) WebSocket communication
- c) CSV downloads
- d) Batch APIs

Correct Answer: b. WebSocket communication

Detailed Solution: Refer to lecture 15.

QUESTION 8 : [1 marks]

Why is server-side scripting preferred for database access in web applications?

- a. It reduces network latency
- b. It exposes database credentials to users
- c. It centralizes logic and improves security
- d. It replaces the need for APIs

Correct Answer: c. It centralizes logic and improves security

Detailed Solution: Refer to lecture 15.

QUESTION 9 : [2 mark]

Match the following SQL Clauses & Their Roles:

Clauses	Roles
P. WHERE	i. Filters rows after aggregation
Q. HAVING	ii. Sorts the query output
R. GROUP BY	iii. Filters rows before aggregation
S. ORDER BY	iv. Groups rows based on column values

Choose the correct option:

- a. P–iii, Q–i, R–iv, S–ii
- b. P–i, Q–iii, R–ii, S–iv
- c. P–iv, Q–i, R–iii, S–ii
- d. P–iii, Q–iv, R–i, S–ii

Correct Answer: a. P–iii, Q–i, R–iv, S–ii

Detailed Solution: Refer to lecture 14.



QUESTION 10 : [1 mark]

The SQL constraint that enforces a logical condition such as `area_sq_m > 50` is called _____.

- a. UNIQUE constraint
- b. DEFAULT constraint
- c. CHECK constraint
- d. NOT NULL constraint

Correct Answer: c. CHECK constraint

Detailed Solution: Refer to lecture 12.

QUESTION 11 : [1 mark]

Which situation requires the use of the HAVING clause instead of the WHERE clause?

- a. Filtering rows based on a non-numeric column
- b. Filtering rows before grouping
- c. Filtering results based on an aggregate condition
- d. Sorting query output

Correct Answer: c. Filtering results based on an aggregate condition

Detailed Solution: Refer to lecture 13.

QUESTION 12 : [1 mark]

Which constraint ensures referential integrity between two related tables?

- a. UNIQUE
- b. PRIMARY KEY
- c. CHECK
- d. FOREIGN KEY

Correct Answer: d. FOREIGN KEY

Detailed Solution: Refer to lecture 12.

QUESTION 13 : [1 mark]

The constraint that automatically assigns a value to a column when no value is provided during insertion is _____.

- a. CHECK
- b. UNIQUE
- c. DEFAULT
- d. NOT NULL

Correct Answer: c. DEFAULT



Detailed Solution: Refer to lecture 12.

QUESTION 14 : [2 marks]

Match the following Data Access & Deployment Concepts:

Digital Participation	Description
P. REST API	i. Client-driven data querying allowing flexible response structure
Q. ORM	ii. Database offered as a fully managed cloud service
R. GraphQL	iii. Object-based interaction with relational databases
S. Managed Cloud Database	iv. Stateless request-response communication over HTTP

Choose the correct option:

- a. P-iv, Q-iii, R-i, S-ii
- b. P-iii, Q-iv, R-ii, S-i
- c. P-i, Q-ii, R-iv, S-iii
- d. P-iv, Q-i, R-iii, S-ii

Correct Answer: a. P-iv, Q-iii, R-i, S-ii

Detailed Solution: Refer to lecture 15.



QUESTION 15 : [2 mark]

A city traffic department maintains a database to analyze traffic flow and road safety across different road segments. The database consists of the following two tables:

Table 1: Road_segments

Column Name	Data Type	Description
road_id	INT	Unique ID for each road segment
road_name	VARCHAR	Name of the road
area	VARCHAR	City zone (CBD / Residential / Industrial)
speed_limit	INT	Speed limit in km/h

Table 2: Traffic_observations

Column Name	Data Type	Description
obs_id	INT	Observation ID
road_id	INT	References road_segments(road_id)
avg_speed	FLOAT	Average vehicle speed (km/h)
traffic_volume	INT	Vehicles per hour
obs_date	DATE	Date of observation

Which of the following SQL components are required to compute the average traffic volume per area? (Select all that apply)

- a. JOIN
- b. GROUP BY
- c. AVG()
- d. HAVING

Correct Answers: a, b, & c

Detailed Solution: Refer to lecture 14.

JOIN operation is used on area from the road_segments table and traffic_volume from the traffic_observations table.

AVG() aggregate function is used to calculate the average traffic volume

GROUP BY is required to perform this aggregation separately for each area.



QUESTION 16 : [1 mark]

From the above given database, Which SQL query correctly identifies road segments that have more than one traffic observation recorded?

- a.

```
SELECT road_id
FROM traffic_observations
WHERE COUNT(obs_id) > 1;
```
- b.

```
SELECT road_id
FROM traffic_observations
GROUP BY road_id
HAVING COUNT(obs_id) > 1;
```
- c.

```
SELECT road_id
FROM road_segments
GROUP BY road_id
HAVING COUNT(obs_id) > 1;
```
- d.

```
SELECT road_id
FROM traffic_observations
HAVING COUNT(obs_id) > 1;
```

Correct Answer: b

```
SELECT road_id
FROM traffic_observations
GROUP BY road_id
HAVING COUNT(obs_id) > 1;
```

Detailed Solution: Refer to lecture 14.

The **FROM traffic_observations** clause specifies the table containing all traffic observation records.

The **SELECT road_id** clause defines the required output column.

The **GROUP BY road_id** clause groups observations for each road segment

COUNT(obs_id) computes the number of observations per group.

Since filtering is based on an aggregate value, the **HAVING COUNT(obs_id) > 1** clause is used.

*****END*****